Measures of Location

The tendency of clustering the values around the centre of the series is usually called central tendency. And it's numerical measure are called measures of central-location / central tendency.

Characteristics of an ideal measure of location:

i) It should be rigidly defined.
ii) It should be readily understandable and easy to calculate.
iii) It should be based upon all the observation.
iv) It should be suitable for further algebric treatments.
v) It should be affected as small as possible by sampling fluctuation.

Different important measure of location:

1) Arithmatic mean or mean 2) Geometric mean 3) Harmonic mean
4) Median 5) Mode.

1) Arithmatic mean (A.M):

Sum of all observations divided by the number of observation.

A.M. of n ungroup observation $x_1, x_2, x_3 \dots x_n$ is given by

$$\bar{x} = \frac{x_1 + x_2 + x_3 \dots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- **Short cut method:** If (x_i) and f_i are large the calculation of $\bar{x}$ by usual formula is time consuming. This can be simplified by defining a new variable,

$$u_i = \frac{x_i - A}{h} \quad \left[\begin{array}{l} x_i \to \text{mid value of class interval}, \\ A \to \text{arbitary number called origin} \end{array}\right. \quad h \text{ is scale} \to \text{generally length of class interval}$$

$$\bar{x} = \frac{\sum x_i}{n} = \frac{\sum h u_i + A}{n} = h\bar{u} + A \quad \left| \left(\bar{u} = \sum f_i u_i / N\right)\right.$$

- **Properties**

1) Sum of the deviations of the values of the variable from its arithmetic mean is zero.

$$\sum_{i=1}^{n} f_i (x_i - \bar{x}) = \sum_{1}^{n} f_i x_i - \sum f_i \bar{x} = n\bar{x} - n\bar{x} = 0 \quad \left| \begin{array}{l} \bar{x} \text{ is the mean of } x_i \text{ with} \\ f_i \text{ frequency} \end{array}\right.$$

2) The A.M of a set of N constant observations A is A.

i.e. $\bar{x} = \frac{\sum_{i=1}^{N} A}{N} = \frac{NA}{N} = A.$

3) The sum of the squares of the deviations of a set of values from the A.M is minimum.

Let, $\sum_{i=1}^{n} f_i (x_i - \phi)^2 = \sum_{i=1}^{n} f_i (x_i - \bar{x} + \bar{x} - A)^2$

$$= \sum_{i=1}^{n} f_i \{(x_i - \bar{x}) + (\bar{x} - A)\}^2$$

$$= \sum_{i=1}^{n} f_i (x_i - \bar{x})^2 + \sum_{i=1}^{n} f_i (\bar{x} - A)^2 + \sum f_i \cdot 2 \cdot (x_i - \bar{x})(\bar{x} - A)$$

$$= \sum_{i=1}^{n} f_i (x_i - \bar{x})^2 + n(\bar{x} - A)^2 \quad \because [(x_i - \bar{x}) = 0]$$

$$= \sum f_i (x_i - \bar{x})^2 + n(\bar{x} - A)^2$$

here $n(\bar{x} - A)^2 \to$ always +ve

Thus, $\sum_{i=1}^{n} f_i (x_i - \bar{x}) < \sum_{i=1}^{n} f_i (x_i - A)^2$ (proved)

4) If $\bar{x}_i (i = 1, 2, 3, \dots k)$ are the k component series of size $n_i \{i = 1, 2 \dots k\}$ respectively, then the mean $\bar{x}$ of the composite series,

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + \dots + n_k \bar{x}_k}{n_1 + n_2 + n_3 \dots n_k}$$

• Merit of AM
a) It is rigidly defined and easy to calculate.
b) It takes all the observation into account.
c) It is less affected by sampling fluctuation.

• Demerit of AM
a) It is affected by extreme values.
b) It is impossible to calculate if the extreme classes of the frequency distribution are open.

Remarks

1) $\sum_{i=1}^{n} a x_i = a x_1 + a x_2 + \dots + a x_n = a(x_1 + x_2 \dots + x_n) = a \sum x_i$

here a is constant

2) $\sum_{i=1}^{n} (a x_i + b y_i) = a \sum_{i=1}^{n} x_i + b \sum_{i=1}^{n} y_i$

3) $\sum_{i=1}^{n} x_i f_i = x_1 f_1 + x_2 f_2 + \dots + x_n f_n$

4) $\sum_{i=1}^{n} c = c + c + c \dots + c = nc$

5) $\sum_{i=1}^{n} (x_i + c) = \sum x_i + \sum_{i=1}^{n} c = \sum x_i + nc$

6) $\sum_{i=1}^{n} x_i^2 = x_1^2 + x_2^2 + \dots x_n^2$

7)

2. Geometric Mean (G.M)

The G.M. of a set of n non-zero positive observations is the nth root of their product.

Let, $x_1, x_2 \dots x_n$ be n non-zero positive observation in a series of data,

$$G = (x_1 \cdot x_2 \dots x_n)^{\frac{1}{n}}$$

Simplified by taking logarithm

$$\log G = \frac{1}{n} \log (x_1 \cdot x_2 \cdot \dots x_n)$$

$$= \frac{1}{n} \sum_{i=1}^{n} \log x_i$$

Merit:
- AM merits
- It gives comparitively more weight to small observations

Demerit:
- It is undefined for negative observation and zero
- It is difficult to understand and to calculate for a person with less mathmetical knowledge.

3. Harmonic Mean (H.M)

The H.M of a set of n non-zero observations observations $x_1, x_2 \dots x_n$ in a series is the reciprocal of the A.M of the reciprocal of the given value.

$$HM = \frac{1}{\frac{1}{n} \sum \frac{1}{x_i}} = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

merit:
- It is rigidly defined
- It takes all the observation into account
- It is not affected much by small observation

demerit:
- It is not easy to understand and difficult to calulate
- It is impossible to calculate if the extreme classes of the frequency distribution are open.

Uses:

AM: Generally uses this mean.

GM: i) For the construction of the index numbers,
ii) If the set of the quantities are in geometric progression.
iii) If the observation is ratio and percentage, the GM is the appropriate averages.

HM: i) The harmonic mean is often appropriate when we are dealing with rates, speeds and prices.

Relationship among AM, GM, HM:

1) $AM \geq GM \geq HM$

Let the two non zero positive observation x_1 and x_2

$AM = \frac{x_1+x_2}{2}$ $\quad GM = (x_1 \cdot x_2)^{\frac{1}{2}} = \sqrt{x_1 x_2}$ $\quad HM = \frac{1}{\frac{1}{2}\left(\frac{1}{x_1}+\frac{1}{x_2}\right)} = \frac{2x_1x_2}{x_1+x_2}$

- (any square will always non negative)

✓ $(\sqrt{x_1}-\sqrt{x_2})^2 \geq 0 \Rightarrow (x_1+x_2-2\sqrt{x_1x_2}) \geq 0 \Rightarrow x_1+x_2 \geq 2\sqrt{x_1x_2} \Rightarrow \frac{x_1+x_2}{2} \geq \sqrt{x_1x_2}$ $\because AM \geq GM$

✓ $(x_1+x_2) > 2\sqrt{x_1x_2} \Rightarrow \frac{2\sqrt{x_1x_2}}{x_1+x_2} \leq 1 \Rightarrow \frac{2x_1x_2}{x_1+x_2} < \sqrt{x_1x_2}$ $\because HM \leq GM$

So, $AM \geq GM > HM$ (proved)

2) $AM \cdot HM = GM^2$ $\quad \frac{x_1+x_2}{2} \times \frac{2x_1x_2}{x_1+x_2} = x_1x_2 = \left(\sqrt{x_1x_2}\right)^2 = GM^2$

4) median: The middle most observation when the observation are arranged in order of magnitude.

- For ungroup datas
  a) n is odd, the median will be $\left(\frac{n+1}{2}\right)$th observation
  b) n is even, the median will be arethmetic mean of $\left(\frac{n}{2}\right)$th and $\left(\frac{n}{2}+1\right)$th observation

- For group data,

$$Me = L + \frac{\frac{n}{2}-F}{f} \times C$$

where,
L → The lower limit of median class ($\frac{n}{2}$ → in which class)
n → no. of observation
F → cumulative freq. of pre median class
f → frequency of median class
C → length of median class

⊛ merit: i) It is not affected by extreme values.
2) It can be calculated from frequency distribution with open end
3) It is rigidly defined.
4) Easy to understand and calculate.

⊛ Demerit: i) In case of the even no. of observations, median cannot be defined exactly.
2) It is not based on all observation
3) It is affected much by sampling fluctuation.

5. Mode

Mode is that observation of the variable for which the frequency is maximum.

$$M_o = L + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} \times h$$

L - Lower limit of modal class
f_0 - frequency of modal class
f_1 - frequency of pre-modal class
f_2 - frequency of post-modal class
h - length of modal class.

merit
1) Easy to understand and calculate.
2) It is not affected by extreme value.
3) It can be calculated from frequency distribution with open class.

demerit
1) Mode is not defined clearly in case of bi-modal or multimodal distribution.
2) It is not based on all observation
3) It is affected to a great extent by sampling fluctuation.

Other Measures of Location

Quantile → (Quantile, Deciles and Percentiles)

Quantile: Quantiles are those values in a series which divide the total frequency into number of equal parts when the series is arranged in order of magnitude.

some important quantile (quantile, deciles, percentile)

- Quantile: Quantile are those values which devide the total frequency into four equal part. Thus there are 3 quantile denoted by Q_1, Q_2, Q_3.
- Deciles: Ten equal part I denoted by $D_1, D_2 \dots D_9$.
- percentile: The p-th percentile is a value that divides the dataset in two parts. Approximately p% of the items values are less than p-th percentile, and approximately (100-p)% of the item are greater values.

- Quantile, $Q_i = L_i + \frac{\frac{iN}{4} - F_i'}{f_i} \times h$; $i = 1, 2, 3$
- Decile → $D_j = L_j + \frac{\frac{jN}{10} - F_j'}{f_j} \times h$; $j = 1, 2 \dots 9$
- Percentile, $P_k = L_k + \frac{\frac{kN}{100} - F_k'}{f_k} \times h$; $k = 1, 2 \dots 99$

$Me = Q_2 = P_{50} = D_5$

F_i → Cumulative frequency of pre a class
f_i → frequency of that class

Emperical Relationship among A.M, median and Mode

1) Moderately assymetrical distribution, (Anithmatic mean - Mode) = 3 (Arithmatic mean - Median)
2) positively skew distribution, mode < median < mean
3) Negatively skew distribution, mode > median > mean
4) moderately skew distribution Mode = 3 median - 2 mean
5) Symetrical distribution, all coincide (mean, median, mode) same.

